09 · The words of genetics

The words of this track, as it says them. Bold brown is the examiner.

The words of genetics. Her student-marked quiz has you match sixteen of them.

Define these terms: homozygous, heterozygous, allele, dominant, recessive, genotype, phenotype, autosomal, X-linked, locus, chromosomes, genome, autosome, sex chromosome, variation, and inheritance.

Two of her checklist points are nothing but definitions, and she re-asks the same dozen words across eight different quizzes. She never wants an essay about a term. She gives you the definition and wants the word, or gives you the word and wants the definition. So these are the cheapest marks in the module. Every definition here is her quiz key's.

Start with the pair everyone mixes up. Homozygous: the alleles of the genotype are the same. Big B big B, or little b little b. Homo means same. Heterozygous: the alleles of the genotype are different. Big B little b. Hetero means different.

An allele is a different version of a gene. Her example: a diseased gene versus a non-diseased gene. Her other wording: alternative forms of genes are called alleles.

Dominant: it will always appear in the phenotype, if it's there. It suppresses the expression of the other allele, and it's written with a capital letter. Recessive: it will only appear in the phenotype if no other alleles are present, so only when there are two copies. It's written in lower case.

Genotype: the alleles that have been inherited by an individual. The letters. Phenotype: how the traits are expressed and physically seen in the person, like freckles you can see. Her other wording: the expression of genes is called the phenotype.

Autosomal: when a disease gene is found on chromosomes one to twenty-two. X-linked: when a disease gene is found on the X chromosome, and it generally shows up in greater numbers in males.

Locus: where a gene is located on a chromosome. Think of it as the gene's address.

Chromosomes: a length of DNA that has a collection of genes on it. They're found in pairs. Genome: all the DNA found within the nucleus. Autosome: a chromosome numbered one to twenty-two. Sex chromosome: the X or the Y.

Variation: differences in the genes found in populations. Inheritance: how genes are passed from parents to children.

Five more words her other quizzes use. A gene: a segment of DNA that codes for a trait or a protein. A carrier: a heterozygous person who can pass on a recessive condition without having it. Heredity: the passing of traits from parents to children. Genetics: the study of how traits are inherited through the interactions of alleles. And a phenocopy: a phenotype that looks like a genetic condition, but is caused by the environment, with no genetic basis.

The traps. Autosomal describes where a disease gene sits; an autosome is the chromosome itself. The genome is all of the DNA; a gene is one segment of it. And a chromosome isn't a gene: it carries many of them.

Once more. Sixteen words.

Homozygous: the alleles are the same. Heterozygous: they're different. Allele: a version of a gene. Dominant: always appears in the phenotype. Recessive: appears only if no other allele is present. Genotype: the alleles inherited. Phenotype: the traits you can see. Autosomal: a disease gene on chromosomes one to twenty-two. X-linked: on the X, and more common in males. Locus: where a gene sits on a chromosome. Chromosomes: lengths of DNA carrying genes, in pairs. Genome: all the DNA in the nucleus. Autosome: a chromosome numbered one to twenty-two. Sex chromosome: the X or the Y. Variation: differences in genes across populations. Inheritance: how genes pass from parents to children.